

MARKOSE TSEGAYE

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Education

George Mason University

Bachelor of Science in Computer Science (Accelerated M.S. Track)

Dec 2027

Fairfax, VA

Experience

Software Engineer

May 2025 - July 2025

The Dumpster Masters | Local Waste Management Client

Rockville, MD

- Delivered a React Native MVP for a local trash compaction company, supporting **3** pilot drivers and **100+** jobs weekly.
- Designed a route planning system using mapping APIs and geospatial data, improving operational efficiency by **35%** through performance optimization.
- Implemented a secure Express.js REST API with **20+** endpoints supporting authentication, job scheduling, billing, and admin workflows, handling **200+** daily requests with PostgreSQL-backed transactional integrity.
- Engineered backend support for Dumpster Masters' appointment scheduling, including time-slot locking, cron-driven cleanup jobs, and race-condition protection to prevent double bookings under concurrent booking requests.
- Met weekly with client's product lead to review progress and prioritize features, reducing mid-sprint changes by **70%**.

Software Engineer

July 2025 - Sept 2025

Forkcast | Predictive Restaurant Platform

Remote

- Built the backend for a restaurant demand-forecasting platform, processing POS datasets for predictive insights.
- Owned backend data intake and processing pipelines while collaborating with a **5-member** agile team.
- Developed Express and FastAPI ingestion APIs backed by BullMQ workers to asynchronously process **300K+** rows of POS datasets, enabling near real-time inventory trends and recommendations without blocking client requests.
- Iterated on architecture and feature scope through continuous feedback from startup founders and restaurant interviews, translating product discussions into actionable backend features.

Projects

AlgoMinds (*Algominds.net*) | *Typescript, Next.js, Express.js, WebSockets, Redis*

March 2025 – Present

- Built and scaled AlgoMinds, a real-time multiplayer coding platform supporting **100+** concurrent users.
- Led backend development while collaborating with another engineer who handled authentication and frontend integration, ensuring seamless interaction between real-time and user systems.
- Designed and implemented a distributed systems pipeline using Redis queues and asynchronous processing to execute user-submitted code, reducing execution latency by **40%** and preventing system bottlenecks.
- Deployed backend services to Fly.io and Azure, integrating centralized logging with Winston and Sentry error tracking.
- Reduced unhandled runtime failures by **40%** through structured error handling and proper monitoring.

DebateGuard | *Typescript, Next.js, React, Git, Node.js*

July 2025 - Sept 2025

- Built DebateGuard, a full-stack Next.js/TypeScript web app for structured debate interaction in under **48** hours.
- Implemented key React components, modeling debate states, and user interactions with clear, maintainable TypeScript.
- Contributed to system design, feature planning, and iterative development through collaborative Git workflows.

Leadership

Event Coordinator - ColorStack GMU

Oct 2025 – Present

George Mason University

Fairfax, VA

- Planned technical workshops and networking events supporting over **100+** underrepresented CS students at GMU.
- Worked with industry partners to organize events that connect members with internship and mentorship opportunities.
- Managed event logistics, budgets, and promotion efforts to ensure successful turnout and professional execution.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C, SQL, HTML/CSS

Frameworks: React, React Native, Next.js, Express.js, FastAPI, Django

Data & Infrastructure: Node.js, Redis, BullMQ, Postgres, Prisma ORM, REST APIs, WebSockets

Cloud & Tooling: Docker, Linux, Azure, AWS, Fly.io, Vercel, GitHub, Jest, Sentry, VS Code,